

Network Troubleshooting - Hardware

Troubleshooting an internet outage in a restaurant is essential to ensure a smooth operation, as many restaurant functions rely on a stable internet connection, such as processing payments, managing orders, and providing Wi-Fi for customers. Here's a step-by-step guide to help you identify and resolve internet issues:

This article covers mostly physical steps with hardware that can be taken by anyone on location. For more in depth troubleshooting anything involving software, contact a onePOS tech via our support line.

Preliminary

Here is some information that should be determined before you begin network troubleshooting.

- Have there been any recent changes to any network configuration physically or through software?

We need to know if the location has possibly gotten a new router or possibly unplugged something.

Specific Questions:

- What changed between now and when internet went out?
- Did we install a new modem, router, or switch?
- Did we recently unplug or reseat any network connections?

- Is Terminal 1 offline or is it all terminals?

This can help determine if there is a network issue within the onePOS network or within the entire location's network

Specific Questions:

- Is there is network on the other devices at the location? (phones, TVs, etc.)
- If there is network on other devices, what network are they connected to?

Reasoning:

- If everything at the location is offline, then likely will need assistance directly from the Internet Service Provider (ISP)

- Have there been any recent changes to any network configuration physically or through software?

We need to know if the location has possibly gotten a new router or possibly unplugged something.

Specific Questions:

- What changed between now and when internet went out?
- Did we install a new modem, router, or switch?
- Did we recently unplug or reseal any network connections?

Reasoning:

- Knowing what changes were made may give an idea of where to start looking in troubleshooting process
- If a new modem or router is installed, terminal software reconfiguration will need to be done ([Changing Router or ISP](#))
- If you recently made this change to a new router or isp, definitely let a onePOS support tech know.

- Do the Ethernet cables have lights on both ends?

We need to be sure that ethernet cables plugged into the terminals on one end and either wall / router / or switch on the other end have typically green and orange flashing lights for any and all connections along the path.

Specific Questions:

- Are there lights flashing on the Ethernet cable when it is plugged into the port?
- Are you able to trace this cable to the other end and determine if there are lights there as well?

-What does T1 network status say?

Terminal software and launcher window need to be completely closed out so that only the windows desktop is viewable.

Specific Questions:

- Is the network symbol a globe?
- If it is instead a computer symbol, does it have a yellow triangle or red x?

Reasoning:

- A **globe** () indicates the device is connected to the network, but the network is not connected to the internet. May indicate ISP or router issue.
- **Yellow triangle** means there is some type of connection but there is an issue with it.

❗ Sometimes just trying network troubleshooter resolves this or will better explain the problem.

- **Red X** means there is no connection being made, meaning the terminal is not detecting anything plugged into the ethernet port. May indicate ethernet cable needs replacement especially if we checked and saw there are no lights on it.

-Where is network equipment located?

We'll need to know how a location's network is set up, specifically if there are multiple routers or switches and if there are a separate or combined modem + router.

Specific Questions:

- Do the network cables go from the terminal direct to the switch or does it go into the wall?
- Is all network equipment located in the same place
- Do we know if there are multiple switches or routers?
- Do we know if the router and modem are combined or separate? (hint: majority are combined, they look the same)

Reasoning:

- If ports go directly to switch, it is easiest to determine which ports on the switch are for what terminals / equipment if you need to merely swap a port
- We need to know where network equipment is so that we can verify its status.
- If there are multiple routers, we need to know which one is specific to onePOS.

Step By Step

1. Check for Power and Connections:

- Ensure that your modem, router, and any network equipment are powered on and connected to electrical outlets.
- Verify that all cables (power, Ethernet, and phone lines) are securely plugged in on both ends.
- Verify that all ethernet cables have lights on both ends and replace them if not.

2. Check Network Switch:

❗ Sequence of network equipment: **Modem -> Router -> WTI (optional) -> Switch**

- If you are tracing cables, your terminal cables will either go directly to a switch or into the wall that will route to the switch

- A switch looks like a box with a bunch of ethernet ports on it.



- You'll want to verify whether the ports on this switch , empty or not, have flashing lights. This will indicate communication being made.

- Some switch's have their cables labeled, most do not.



- If you are having issues with just one terminal, try to determine the port that is not lit up and try swapping over to another open port on the switch. Always try to make note of the original location (take picture before making changes) if you do this, especially on larger switches

- If you swap ports and are still experiencing issues, move to the next step.

3. Restart Network Equipment

- In order you'll want to attempt these one by one, switch, motom, router. If the first step works, stop there. If it doesn't, move on to the next item in the list

- Turn off the equipment via unplugging the power cable and **then waiting for about 30 seconds before turning them back on.**

❗ It is important to wait these 30 seconds to allow the device to fully restart.

- Allow a few minutes for the devices to initialize and reconnect to the internet.

❗ - Be sure to monitor the terminals along with network equipment to see if any changes are occurring during your processes.

4. Check Network Equipment

- For each device, trace to see if there are cables other than the power cable (ex: another ethernet cable) plugged into the equipment

- **You'll want to make sure that every cable along the line for each piece of equipment is connected securely and has lights on it.**

- Additionally you can ensure the outlets being used by each piece of equipment has power and check the fuse box for blown or reset switches if necessary.

5. Check ISP Status:

- Visit your Internet Service Provider's website or call their customer support line to check if there are known outages or maintenance in your area. On the physical router itself you can typically find the brand logo and can search for "X internet service provider support number" if you do not have it already.

❗ Remember that troubleshooting an internet outage can be a process that may involve multiple steps and potentially some downtime. Your goal is to identify and resolve the issue as quickly as possible to minimize disruptions to your restaurant's operations.